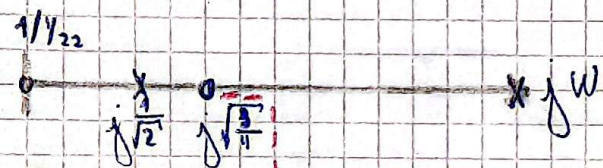


$$5) a) \frac{V_2}{V_1} = \frac{S^2 + 1}{2S^2 + 1} = \frac{-Y_{21}}{Y_{22}}$$

Plantamos como polinomio auxiliar de Y_{22} a: $S(S^2 + \frac{3}{4})$

$$Y_{22} = \frac{2S^2 + 1}{S(S^2 + \frac{3}{4})}$$



$$Z_2 = \frac{1}{Y_{22}} = K_{\infty} \cdot S$$

$$K_{\infty} = \frac{S(S^2 + \frac{3}{4})}{S(2S^2 + 1)} \Big|_{S^2 \rightarrow \infty} = \frac{1}{4}$$

$$Z_2 = \frac{S(S^2 + \frac{3}{4})}{2S^2 + 1} - \frac{1}{4} \cdot S = \frac{S(S^2 + \frac{3}{4}) - \frac{1}{4}S(2S^2 + 1)}{2S^2 + 1}$$

$$Z_2 = \frac{S^3 + \frac{3}{4}S - \frac{1}{2}S^3 - \frac{1}{4}S}{2S^2 + 1} = \frac{\frac{1}{2}S^3 + \frac{1}{2}S}{2S^2 + 1} \rightarrow Y_2 = \frac{2S^2 + 1}{\frac{1}{2}S(S^2 + 1)}$$

NOTA

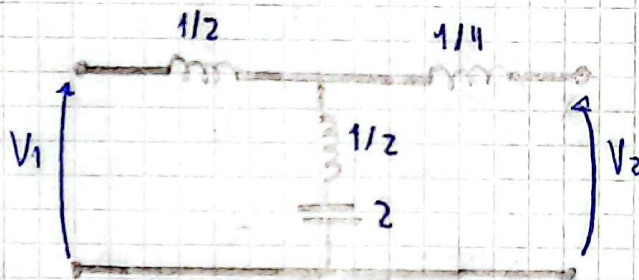
$$Y_4 = Y_2 - \frac{2K_1 S}{S^2 + 1} \quad ; \quad 2K_1 = \lim_{S^2 \rightarrow -1} \frac{(2S^2 + 1) \cdot (S^2 + 1)}{\frac{1}{2}S(S^2 + 1) \cdot S} = 2$$

$$Y_4 = \frac{2S^2 + 1}{\frac{1}{2}S(S^2 + 1)} - \frac{2S}{S^2 + 1} = \frac{2S^2 + 1 - S^2}{\frac{1}{2}S(S^2 + 1)} = \frac{S^2 + 1}{\frac{1}{2}S(S^2 + 1)} = \frac{1}{\frac{1}{2}S}$$

$$Z_4 = \frac{1}{\frac{1}{2}S}$$

$$Z_6 = Z_4 - K_\infty \cdot S \quad ; \quad K_\infty = \lim_{S \rightarrow \infty} \frac{\frac{1}{2}S}{S} = \frac{1}{2}$$

$$Z_6 = \frac{1}{2}S - \frac{1}{2} \cdot S = 0$$



Verificación desde los parámetros impedancia:

$$Z_{12} = Z_{21} = \frac{S}{2} + \frac{1}{2S} = \frac{S^2 + 1}{2S}$$

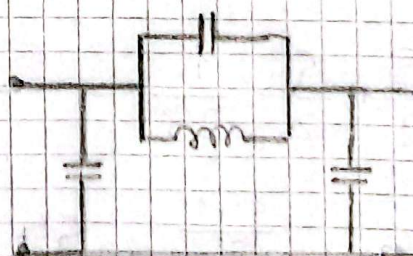
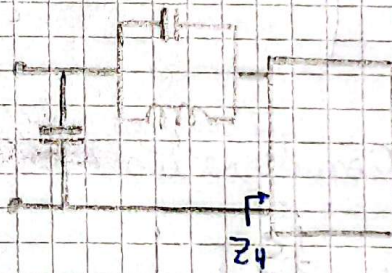
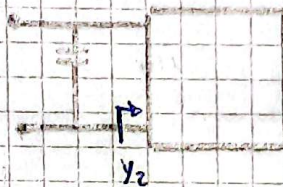
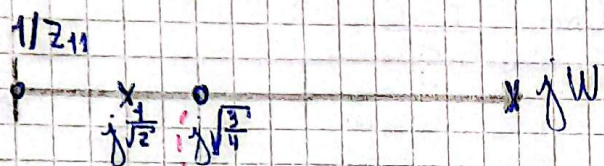
$$Z_{11} = \frac{S}{2} + \frac{S}{2} + \frac{1}{2S} = \frac{S^2 + S^2 + 1}{2S} = \frac{2S^2 + 1}{2S}$$

$$\frac{V_2}{V_1} = \frac{Z_{21}}{Z_{11}} = \frac{\frac{S^2 + 1}{2S}}{\frac{2S^2 + 1}{2S}} = \frac{S^2 + 1}{2S^2 + 1}$$

$$b) \frac{V_2}{V_1} = \frac{S^2+1}{2S^2+1} = \frac{Z_{21}}{Z_{11}}$$

Plantamos como polinomio auxiliar de Z_{11} a: $S(S^2 + \frac{3}{4})$

$$Z_{11} = \frac{2S^2+1}{S(S^2 + \frac{3}{4})}$$



$$y_2 = \frac{1}{Z_{11}} - K_{\infty}' \cdot S \quad ; \quad K_{\infty}' = \frac{S(S^2 + \frac{3}{4})}{S(2S^2+1)} \Big|_{S^2=-1} = \frac{1}{4}$$

$$y_2 = \frac{S(S^2 + \frac{3}{4})}{2S^2+1} - \frac{1}{4}S = \frac{\frac{1}{2}S(S^2+1)}{2S^2+1} \rightarrow z_2 = \frac{2S^2+1}{\frac{1}{2}S(S^2+1)}$$

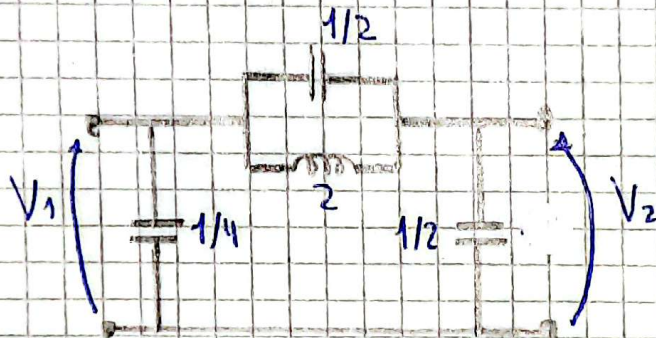
NOTA

$$Z_4 = Z_2 - \frac{2K_1 \cdot S}{S^2 + 1} \quad ; \quad 2K_1 = \lim_{S^2 \rightarrow -1} \frac{(2S^2 + 1) \cdot (S^2 + 1)}{\frac{1}{2}S(S^2 + 1)S} = 2$$

$$Z_4 = \frac{2S^2 + 1}{\frac{1}{2}S(S^2 + 1)} - \frac{2S}{S^2 + 1} = \frac{1}{\frac{1}{2}S} \longrightarrow Y_4 = \frac{1}{2}S$$

$$Y_6 = Y_4 - K_\infty \cdot S \quad ; \quad K_\infty = \lim_{S \rightarrow \infty} \frac{\frac{1}{2} \cdot S}{S} = \frac{1}{2}$$

$$Y_6 = \frac{1}{2}S - \frac{1}{2}S = 0$$



• Verificación desde los parámetros admitancia:

$$-Y_{21} = -Y_{12} = \frac{S}{2} + \frac{1}{2S} = \frac{S^2 + 1}{2S}$$

$$Y_{22} = \frac{S}{2} + \frac{S}{2} + \frac{1}{2S} = \frac{S^2 + S^2 + 1}{2S} = \frac{2S^2 + 1}{2S}$$

$$\frac{V_2}{V_1} = \frac{-Y_{21}}{Y_{22}} = \frac{\frac{S^2 + 1}{2S}}{\frac{2S^2 + 1}{2S}} = \frac{S^2 + 1}{2S^2 + 1}$$